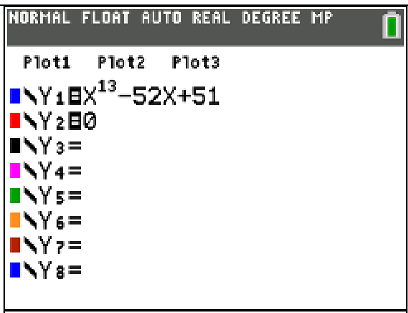
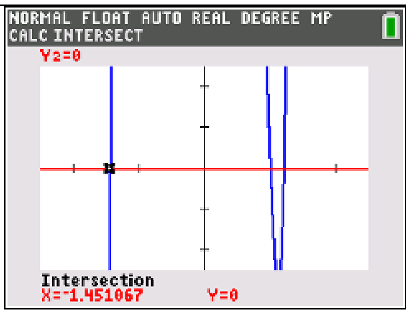
2017 A Level Paper 2 Suggested Solutions

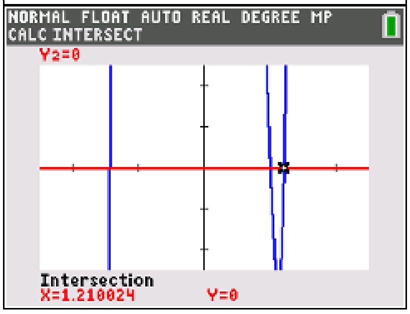
1	**Suggested Solutions**
(i)	$x=\frac{3}{t},\ y=2t$ To find the intersection points between $y=2x$ and curve C, simply substitute $x=\frac{3}{t}$ and $y=2t$ into $y=2x$. $2t=2\left(\frac{3}{t}\right)$ $t^2=3$ $t=\sqrt{3}$ or $t=-\sqrt{3}$ It doesn't matter which is labelled as coordinates of *A* or *B* in this case. When $t=\sqrt{3}$, $x=\frac{3}{\sqrt{3}}=\sqrt{3}$, $y=2\sqrt{3}$, coordinates of point *A* is $\left(\sqrt{3},2\sqrt{3}\right)$ When $t=-\sqrt{3}$, $x=\frac{3}{-\sqrt{3}}=-\sqrt{3}$, $y=-2\sqrt{3}$, coordinates of point *B* is $\left(-\sqrt{3},-2\sqrt{3}\right)$ Length of $AB=\sqrt{\left(\sqrt{3}-\left(-\sqrt{3}\right)\right)^2+\left(2\sqrt{3}-\left(-2\sqrt{3}\right)\right)^2}$ $=\sqrt{\left(2\sqrt{3}\right)^2+\left(4\sqrt{3}\right)^2}$ $=\sqrt{12+48}$ $=\sqrt{60}$ $=2\sqrt{15}$ units
(ii)	$\frac{dx}{dt}=-\frac{3}{t^2}$, $\frac{dy}{dt}=2$ $\frac{dy}{dx}=-\frac{2}{3}t^2$ At $P\left(\frac{3}{p},2p\right)$, $x=\frac{3}{p}\Rightarrow\frac{3}{p}=\frac{3}{t}\quad\therefore t=p$ $y=2p\Rightarrow 2p=2t\quad\therefore t=p$ At the point $P\left(\frac{3}{p},2p\right)$, observe that $t=p$. Hence at $P\left(\frac{3}{p},2p\right)$, $\frac{dy}{dx}=-\frac{2}{3}p^2$ Equation of tangent at $P\left(\frac{3}{p},2p\right)$: $y-2p=-\frac{2}{3}p^2\left(x-\frac{3}{p}\right)$ When $x=0$, $y-2p=-\frac{2}{3}p^2\left(-\frac{3}{p}\right)\Rightarrow y=4p\quad\therefore E(0,4p)$ When $y=0$, $0-2p=-\frac{2}{3}p^2\left(x-\frac{3}{p}\right)\Rightarrow x=\frac{6}{p}\quad\therefore D\left(\frac{6}{p},0\right)$

	Since F is the midpoint of DE, $\therefore$ Coordinates of F are $\left(\dfrac{0+\frac{6}{p}}{2}, \dfrac{0+4p}{2}\right) = \left(\dfrac{3}{p}, 2p\right)$. x-coordinate y-coordinate $x = \dfrac{3}{p} ----(1)$ $y = 2p ----(2)$ From (1), $p = \dfrac{3}{x} ----(3)$ Substitute (3) into (2): $y = 2\left(\dfrac{3}{x}\right) = \dfrac{6}{x}$ Cartesian equation traced by F: $y = \dfrac{6}{x}$ From a set of parametric equations, to find the Cartesian equation, we aim to eliminate the parameter p. Cartesian equation is an equation that involves only variables x and y.

2	**Suggested Solutions**
(i)	Let d be the common difference of the arithmetic progression. $S_{13} = 156$ $\dfrac{13}{2}\left(2(3)+12d\right) = 156$ $d = \dfrac{3}{2}$ For an arithmetic progression, $S_n = \dfrac{n}{2}\left[2a+(n-1)d\right]$
(ii)	$S_{13} = 156$ $\dfrac{3\left(1-r^{13}\right)}{1-r} = 156$ $3\left(1-r^{13}\right) = 156(1-r\pi)$ $1-r^{13} = 52(1-r)$ $1-r^{13} = 52-52r$ $r^{13}-52r+51 = 0$ (shown) For an geometric progression, $S_n = \dfrac{a\left(1-r^n\right)}{1-r}, r \neq 1$ Students are reminded to show sufficient workings for a “shown” question. If the common ratio is 1 and with the first term 3, $S_{13} = 3(13)$ $S_{13} = 39 \neq 156$ (as given by the question) $\therefore r \neq 1$







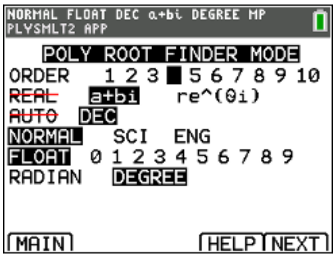
Therefore to solve $r^{13}-52r+51=0$, it is recommended to use the graphical approach.

$r^{13}-52r+51=0$

By GC,

$r=-1.45$ or $r=1.21$ (3 s.f.).

Note: Polysmlt (poly root finder) cannot be used directly here because the highest order of the polynomial is 13 which is out of the range that the GC can support.



(iii) Given that common ratio of the geometric progression is positive,

$r=1.2100$ (5 s.f)

$$3(1.2100)^{n-1}>100\left(3+(n-1)\frac{3}{2}\right)$$

$$3(1.2100)^{n-1}-100\left(3+(n-1)\frac{3}{2}\right)>0$$

For an arithmetic progression, $T_n=a+(n-1)d$

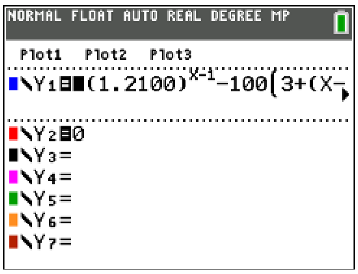
For a geometric progression, $T_n=ar^{n-1}$

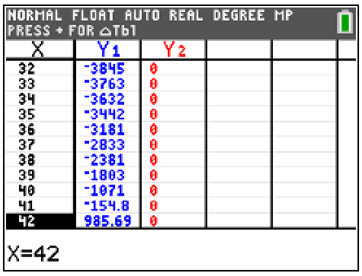
From GC,

When $n=41$, LHS $=-154.8<0$.

When $n=42$, LHS $=985.69>0$.

$\therefore$ Least $n=42$.



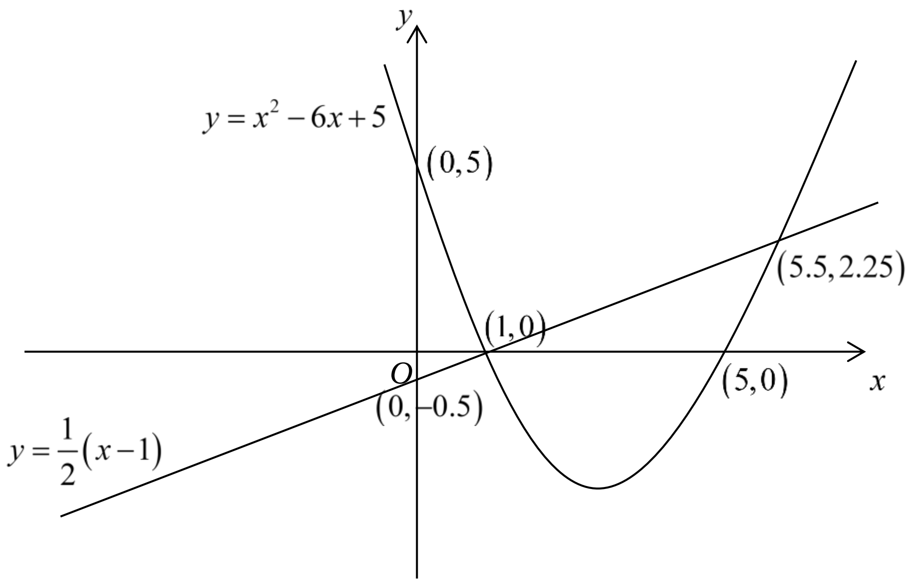


3	**Suggested Solutions**
(a)(i)	$y=\mathrm{f}(x)\to y=\mathrm{f}(2x)$ Replace *x* by 2*x*: Scaling parallel to *x*-axis by factor $\frac{1}{2}$. $(a,0)\to\left(\frac{a}{2},0\right)$ and $(0,b)\to(0,b)$ The points where the curve $y=\mathrm{f}(2x)$ cuts the axes are $\left(\frac{a}{2},0\right)$ and $(0,b)$.
(a)(ii)	$y=\mathrm{f}(x)\to y=\mathrm{f}(x-1)$ Replace *x* by $(x-1)$: Translation of 1 unit in the positive *x*-direction. $(a,0)\to(a+1,0)$ and $(0,b)\to(1,b)$ The point where the curve $y=\mathrm{f}(x-1)$ cuts the axes is $(a+1,0)$.
(a)(iii)	$y=\mathrm{f}(x)\to y=\mathrm{f}(2x-1)$ Replace *x* by $(x-1)$: Translation of 1 unit in the positive *x*-direction. Replace *x* by 2*x*: Scaling parallel to *x*-axis by factor $\frac{1}{2}$. $(a,0)\to(a+1,0)\to\left(\frac{a+1}{2},0\right)$ and $(0,b)\to(1,b)\to\left(\frac{1}{2},b\right)$ The point where the curve $y=\mathrm{f}(2x-1)$ cuts the axes is $\left(\frac{a+1}{2},0\right)$.
(a)(iv)	$y=\mathrm{f}(x)\to y=\mathrm{f}^{-1}(x)$ Reflection in the line $y=x$. The points where the curve $y=\mathrm{f}^{-1}(x)$ cuts the axes are $(0,a)$ and $(b,0)$.
(b)(i)	$a=1$. g is undefined at $x=1$. **Remark:** $x=1$ is the vertical asymptote of the curve $y=\mathrm{g}(x)$

(b)(ii)	$g^2(x)=1-\dfrac{1}{1-\left(1-\dfrac{1}{1-x}\right)}$ $g^2(x)=1-(1-x)=x$ Let $y=1-\dfrac{1}{1-x}$. $y-1=\dfrac{1}{x-1}$ $\dfrac{1}{y-1}=x-1$ (since $y\neq 1$) $x=1+\dfrac{1}{y-1}$ $g^{-1}(x)=1+\dfrac{1}{x-1}$ **Note:** To find $g^2(x)$ and $g^{-1}(x)$ means to find the expressions for $g^2(x)$ and $g^{-1}(x)$. There is no need to find the domains of $g^2(x)$ and $g^{-1}(x)$.
(b)(iii)	$g^2(b)=g^{-1}(b)$ $b=1+\dfrac{1}{b-1}$ $b-1=\dfrac{1}{b-1}$ $(b-1)^2=1$ $b=1\pm 1$ $b=0$ or $b=2$

4	**Suggested Solutions**

(a)



Area $= \int_1^{5.5} \left[\left(\frac{1}{2}x - \frac{1}{2} \right) - \left(x^2 - 6x + 5 \right) \right] \mathrm{d}x$

$= 15.1875$

Upper curve

lower curve

Use GC to evaluate the integral directly as question did not ask for exact area of the plate.

Steps to solve this problem:

1. Sketch both graphs, $y = x^2 - 6x + 5$ and $2y = x - 1$.
2. Find the points of intersection between these two graphs.
3. Using integration to find the area enclosed by these two graphs, i.e using area bounded by the **upper curve** $2y = x - 1$ and the *x*-axis **minus** area bounded by the **lower curve** $y = x^2 - 6x + 5$ and the *x*-axis.

(b) (i)	$$\text{Volume} = \pi\int_0^1 \left(\frac{\sqrt{y}}{a-y^2}\right)^2 \ \mathrm{d}y$$ Note the axis of rotation here. Volume generated when part of the curve is rotated about the ***y*-axis** is $\pi\int_a^b x^2 \ \mathrm{d}y$. Since the rotation is about the ***y*-axis**, the limits will be from $y=0$ to $y=1$. $$= \pi\int_0^1 \frac{y}{\left(a-y^2\right)^2} \ \mathrm{d}y$$ $$= \pi\int_0^1 y\left(a-y^2\right)^{-2} \ \mathrm{d}y$$ $$= \pi\int_0^1 \left(-\frac{1}{2}\right)\underbrace{\left(-2y\right)\left(a-y^2\right)^{-2}} \ \mathrm{d}y$$ **Recall:** $\int \mathrm{f}'(x)\left[\mathrm{f}(x)\right]^n \ \mathrm{d}x = \frac{\left[\mathrm{f}(x)\right]^{n+1}}{n+1} + C, C \in \mathbb{R}$ **Note:** a is a constant greater than 1. $$= \pi\left[\left(-\frac{1}{2}\right)\frac{\left(a-y^2\right)^{-1}}{-1}\right]_0^1$$ $$= \pi\left[\frac{1}{2}\left(a-y^2\right)^{-1}\right]_0^1$$ $$= \frac{\pi}{2}\left[\frac{1}{a-y^2}\right]_0^1 = \frac{\pi}{2}\left(\frac{1}{a-1}-\frac{1}{a}\right)$$ $$= \frac{\pi}{2}\left(\frac{a-(a-1)}{a(a-1)}\right) = \frac{\pi}{2a(a-1)}$$ **Requirement from question:** answer must be given as a single fraction in terms of a and π.
(b) (ii)	$$\frac{\pi}{2b(b-1)} = 4\frac{\pi}{2a(a-1)}$$ There is no need to re-calculate the volume of another container with the equation $x=\frac{\sqrt{y}}{b-y^2}$ as it is rotated the same way as $x=\frac{\sqrt{y}}{a-y^2}$ in part **(i).** Therefore, we can directly used the result $\frac{\pi}{2b(b-1)}$, just by replacing constant a with b. $$b(b-1) = \frac{1}{4}a(a-1)$$ $$\left(b-\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 = \frac{1}{4}a(a-1)$$ $$\left(b-\frac{1}{2}\right)^2 = \frac{1}{4}\left(1+a(a-1)\right)$$ $$b = \frac{1}{2} \pm \sqrt{\frac{1}{4}\left(1+a(a-1)\right)}$$ $$b = \frac{1}{2} \pm \sqrt{\frac{1}{4}\left(1+a(a-1)\right)}$$ $$b = \frac{1}{2}+\frac{1}{2}\sqrt{1+a(a-1)} \quad \text{or} \quad b = \frac{1}{2}-\frac{1}{2}\sqrt{1+a(a-1)}$$ Since $a>1$, $b = \frac{1}{2}-\frac{1}{2}\sqrt{1+a(a-1)} < 0$ $$\therefore b = \frac{1}{2}\left(1+\sqrt{1+a(a-1)}\right) \qquad (\text{since } b \text{ is positive})$$

5	**Suggested Solutions**
(i)	

t	2	3	4	5
$\mathrm{P}(T=t)$	$\frac{6}{9}\left(\frac{5}{8}\right)=\frac{5}{12}$	$\frac{6}{9}\left(\frac{3}{8}\right)\left(\frac{5}{7}\right)\times 2$ $=\frac{5}{14}$	$\frac{6}{9}\left(\frac{3}{8}\right)\left(\frac{2}{7}\right)\left(\frac{5}{6}\right)\times 3$ $=\frac{5}{28}$	$\frac{6}{9}\left(\frac{3}{8}\right)\left(\frac{2}{7}\right)\left(\frac{1}{6}\right)\left(\frac{5}{5}\right)\times 4$ $=\frac{1}{21}$

Explanation:

$$\mathrm{P}(T=2)=\mathrm{P}(R,R)=\frac{6}{9}\left(\frac{5}{8}\right)=\frac{5}{12}$$

$$\mathrm{P}(T=3)=\mathrm{P}(R,Y,R)+\mathrm{P}(Y,R,R)$$
$$=\frac{6}{9}\left(\frac{3}{8}\right)\left(\frac{5}{7}\right)\times 2$$

$$\mathrm{P}(T=4)=\mathrm{P}(R,Y,Y,R)+\mathrm{P}(Y,R,Y,R)+\mathrm{P}(Y,Y,R,R)$$
$$=\frac{6}{9}\left(\frac{3}{8}\right)\left(\frac{2}{7}\right)\left(\frac{5}{6}\right)$$

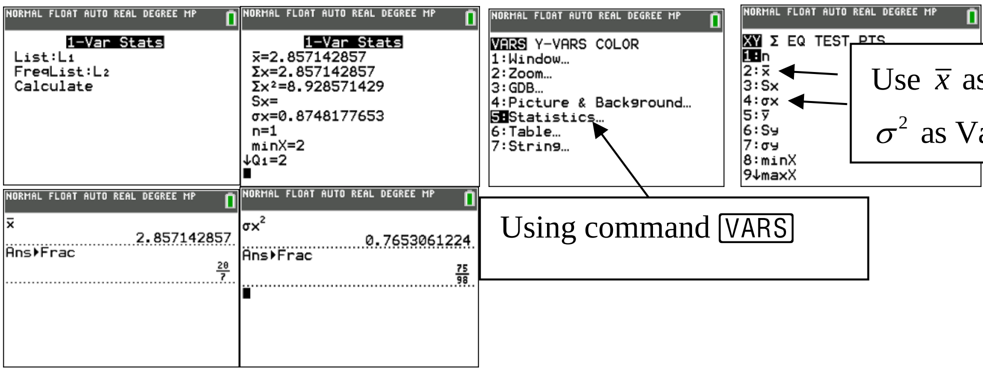
$$\mathrm{P}(T=5)=\mathrm{P}(R,Y,Y,Y,R)+\mathrm{P}(Y,R,Y,Y,R)+\mathrm{P}(Y,Y,R,Y,R)+\mathrm{P}(Y,Y,Y,R,R)$$
$$=\frac{6}{9}\left(\frac{3}{8}\right)\left(\frac{2}{7}\right)\left(\frac{1}{6}\right)\left(\frac{5}{5}\right)\times 4$$

(**Note:** Check that the probabilities sum up to 1.)

(ii) From GC,

$$\mathrm{E}(T)=\frac{20}{7} \quad \text{and} \quad \mathrm{Var}(T)=\frac{75}{98}$$

Students are encouraged to use GC here to find $\mathrm{E}(T)$ and $\mathrm{Var}(T)$.



Use $\bar{x}$ as $\mathrm{E}(T)$ and σ^2 as $\mathrm{Var}(T)$

Using command $\boxed{\text{VARS}}$

(iii) Let X be the number of times, out of 15, that Lee has to take at least 4 counters out of the bag.

$\mathrm{P}(\text{at least 4 counters out of the bag})$

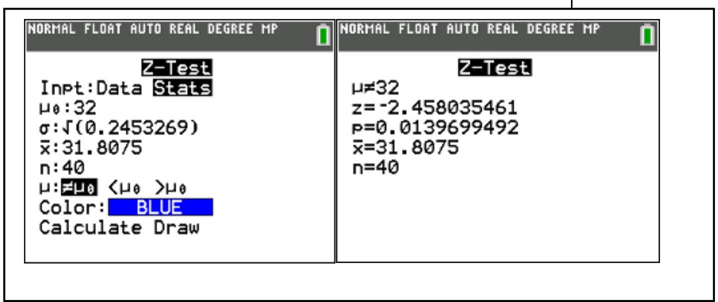
$=\mathrm{P}(T=4)+\mathrm{P}(T=5)$

$=\frac{5}{28}+\frac{1}{21}=\frac{19}{84}$

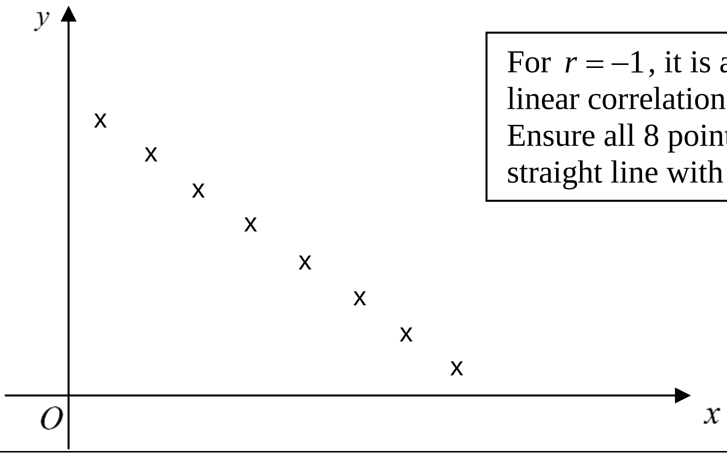
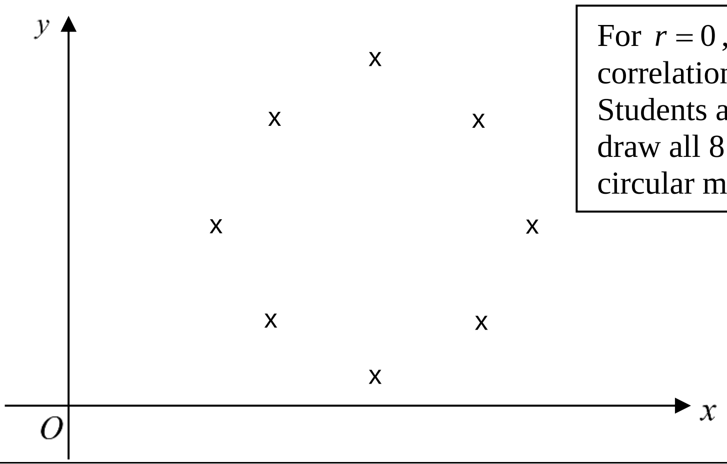
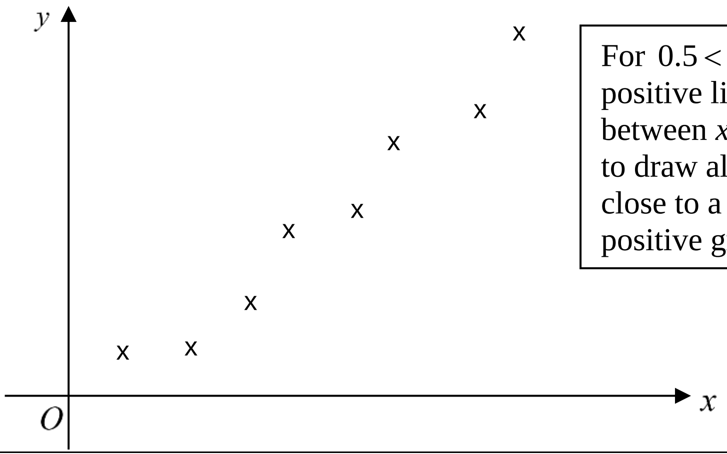
Check conditions FIST for X to follow a binomial distribution.

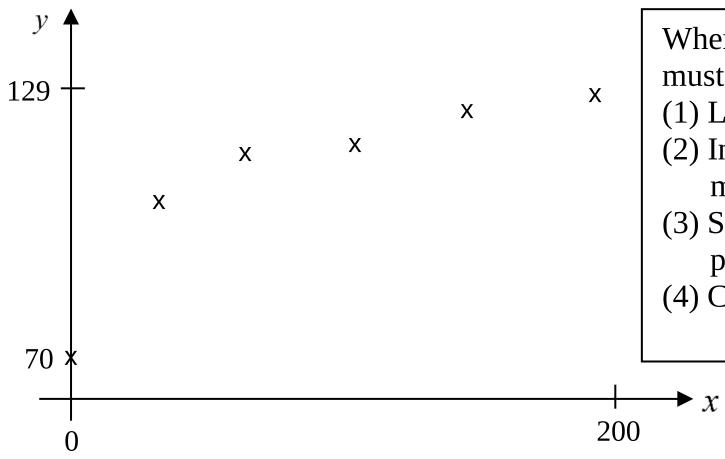
	$X \sim \mathrm{B}\left(15, \frac{19}{84}\right)$ Required probability $= \mathrm{P}(X \geq 5)$ $= 1 - \mathrm{P}(X \leq 4)$ $= 0.238 \quad (3 \text{ s.f.})$

6	Suggested Solutions
(i)	Red family (4 people) Blue family (4 people) Green family (4 people) Yellow family (4 people) Orange family (4 people) Permute 5 units (5 sets of families) Permute 4 people in each of the families Number of ways $= 5! \times (4!)^5 = 955514880$
(ii)	$R_M R_D R_S R_F G_F Y_F O_F B_F B_M B_D B_S$ form 1 unit Remaining people form 9 units Permute $R_M R_D R_S$ Permute $G_F Y_F O_F$ Permute $B_M B_D B_S$ Permute 10 units Permute $R_M R_D R_S$, $G_F Y_F O_F$ and $B_M B_D B_S$ Number of ways $= 10! \times (3!)^3 \times 2! = 1567641600$ It can also be: $B_M B_D B_S B_F G_F Y_F O_F R_F R_M R_D R_S$ hence the need to multiply by 2!.
(iii)	Required probability $= \dfrac{\frac{15!}{15} \times {}^{15}C_5 \times 5!}{\frac{20!}{20}}$ $= \dfrac{1001}{3876}$ or 0.258 To obtain the probability, we need to divide by the total number of ways without restriction, i.e the number of ways to arrange 20 cards in a circular manner is $\dfrac{20!}{20}$. **Slotting method is strongly encouraged here:** If we exclude the 5 father cards, there are 15 cards left. The number of ways to arrange remaining 15 cards in a circular manner is $\dfrac{15!}{15}$. Next, we choose 5 slots out of the 15 slots to slot in the 5 father cards and arrange them to get ${}^{15}C_5 \times 5!$.

7	Suggested Solutions
(i)	A random sample is a selection in which every biscuit bar has an **<u>equal probability of being selected</u>** and whether a biscuit bar is selected **<u>is independent</u>** of any other biscuit bar.
(ii)	Unbiased estimate of population mean, $\bar{x} = \frac{-7.7}{40} + 32$ $= 31.8075$ Remember to add back 32 units to obtain $\bar{x}$. Unbiased estimate of population variance, $s^2 = \frac{1}{39}\left(11.05 - \frac{(-7.7)^2}{40}\right)$ $= 0.2453269$ $= 0.245$ (3 s.f.) Use formula in MF26: $s^2 = \frac{1}{n-1}\left(\sum x^2 - \frac{\left(\sum x\right)^2}{n}\right)$. This formula can still be applied for $s^2 = \frac{1}{n-1}\left(\sum (x-32)^2 - \frac{\left(\sum (x-32)\right)^2}{n}\right)$ because a translation of 32 units does not affect the spread of the data and thus the value of s^2 remains unchanged.
(iii)	Let X be the mass of a randomly chosen biscuit bar (in grams). Let μ be the population mean mass of the biscuit bars (in grams). Let σ^2 be the population variance of the mass of the biscuit bars (in grams^2). $\mu = 32$, $\bar{x} = 31.8075$, $s = \sqrt{0.2453269}$, $n = 40$ $H_0 : \mu = 32$ (claim) $H_1 : \mu \neq 32$ Under H_0, since $n = 40$ is large, by Central Limit Theorem, $\bar{X} \sim N\left(32, \frac{0.2453269}{40}\right)$ approximately. Test Statistic: $Z = \frac{\bar{X} - 32}{\sqrt{\frac{0.2453269}{40}}}$  Level of Significance: 1% Reject H_0 if p-value < 0.01 Under H_0, using GC, p-value $= 0.0140$ (3 s.f.) Since p-value $= 0.0140 > 0.01$, we do not reject H_0 and conclude that there is insufficient evidence, at 1% significance level, that the population mean mass of the biscuit bars is not 32 grams. Hence, the claim is true at 1% level of significance.

(iv)	There is no need to know anything about the population distribution as the sample size, $n = 40$ is large, by Central Limit Theorem, $\overline{X} \sim \mathrm{N}\left(32, \frac{0.2453269}{40}\right)$ approximately.

8	Suggested Solutions
(a) (i)	 For $r = -1$, it is a perfect negative linear correlation between *x* and *y*. Ensure all 8 points drawn lie on a straight line with negative gradient.
(a) (ii)	 For $r = 0$, there is no linear correlation between *x* and *y*. Students are encouraged to draw all 8 points that lie in a circular manner.
(a) (iii)	 For $0.5 < r < 0.9$, there is positive linear correlation between *x* and *y*. Students are to draw all 8 points that lie close to a straight line with positive gradient.

(b) (i)	 When plotting a scatter diagram, you must (1) Label the axes. (2) Indicate the minimum and maximum values on each axes. (3) Show the relative positions of the points. (4) Check that all points are drawn. Model (D) provides the most accurate model of relationship between *x* and *y*. [Good to know: From the scatter diagram, as *x* increases, *y* increases at a decreasing rate, therefor Model D with the equation $y = a\sqrt{x} + b$ provides the most accurate relationship between *x* and *y*.]
(b) (ii)	$y = 4.18211387\sqrt{x} + 74.04787$ $\quad = 4.18\sqrt{x} + 74.0$ (3 s.f.) $r = 0.981$ (3 s.f.)
(b) (iii)	Since $\underbrace{x = 189 \text{ is within the data range of } x}$ and the value of $\underbrace{r = 0.981 \text{ is close to 1,}}$ **Reason 1:** must explicit write $x = 189$ is within the data range of *x*. **Reason 2:** the mention of *r*-value. implying a strong positive linear correlation between $\sqrt{x}$ and *y*. Thus it is reasonable to use model (D) to estimate the value of y when $x = 189$.

<table>
<tr><th>9</th><th>Suggested Solutions</th></tr>
<tr><td>(i)</td><td>1) The probability that a randomly chosen light will be faulty remain constant at 0.08 throughout the sample.
2) Whether a randomly chosen light is faulty is independent of any other lights in the box.</td></tr>
<tr><td>(ii)</td><td>Let X be the number of lights, out of 12, that are faulty.
$X \sim \mathrm{B}(12, 0.08)$
$\mathrm{P}(X \geq 1) = 1 - \mathrm{P}(X = 0)$
$= 0.63233$
$= 0.632$ (3 s.f.)</td></tr>
<tr><td>(iii)</td><td>$\underbrace{\boxed{\begin{matrix}1^{\text{st}}\text{ Box}\\ \mathrm{P}(X \geq 1)\end{matrix}}\ \boxed{\begin{matrix}2^{\text{nd}}\text{ Box}\\ \mathrm{P}(X \geq 1)\end{matrix}}\ \boxed{\begin{matrix}3^{\text{rd}}\text{ Box}\\ \mathrm{P}(X \geq 1)\end{matrix}}\ \ldots\ldots\ \boxed{\begin{matrix}20^{\text{th}}\text{ Box}\\ \mathrm{P}(X \geq 1)\end{matrix}}}_{\boxed{\text{Each box contains at least 1 faulty light}}}$

Required probability $= \left(\mathrm{P}(X \geq 1)\right)^{20}$
$= 0.000104$ (3 s.f.)</td></tr>
<tr><td>(iv)</td><td>Let W be the number of lights, out of 240, that are faulty.

$W \sim \mathrm{B}(240, 0.08)$
$\mathrm{P}(W \geq 20) = 1 - \mathrm{P}(W \leq 19)$
$= 0.45833$
$= 0.458$ (3 s.f.)

There are 20 boxes in one carton and 12 lights in each box. Therefore, the total number of lights is $20 \times 12 = 240$.</td></tr>
<tr><td>(v)</td><td>The events in part (iii) are <u>proper subset</u> of the events in part (iv), thus the answer in part (iv) is greater than the answer in part part (iii).

Further explanation: The following is an example of an event in part (iv) but not in part (iii)

$\boxed{\begin{matrix}1^{\text{st}}\text{ Box}\\ \mathrm{P}(X = 0)\end{matrix}}\ \boxed{\begin{matrix}2^{\text{nd}}\text{ Box}\\ \mathrm{P}(X = 10)\end{matrix}}\ \boxed{\begin{matrix}3^{\text{rd}}\text{ Box}\\ \mathrm{P}(X = 10)\end{matrix}}\ \ldots\ \boxed{\begin{matrix}20^{\text{th}}\text{ Box}\\ \mathrm{P}(X = 10)\end{matrix}}$</td></tr>
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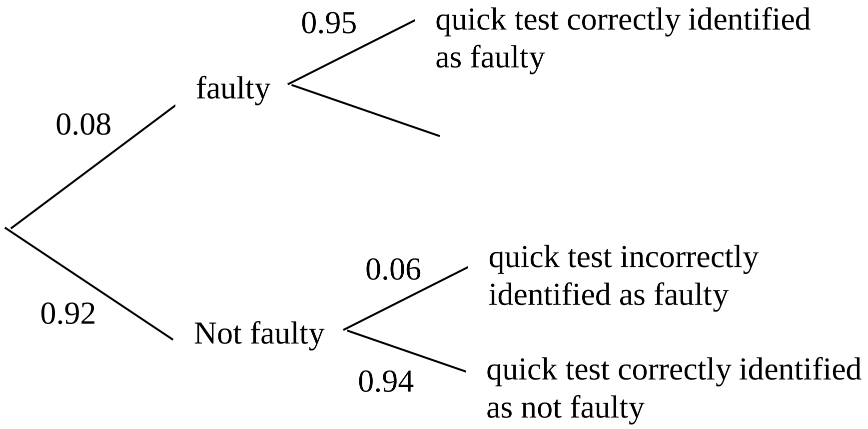
<table>
<tr><td>(vi)</td><td>



Recognise that this question is testing on the concept of conditional probability:
Find the probability that a light identified as faulty by the quick test is not faulty.

Given condition

$\mathrm{P}(\text{not faulty} \mid \text{identified as faulty by quick test})$

$= \dfrac{\mathrm{P}(\text{not faulty} \cap \text{identified as faulty by quick test})}{\mathrm{P}(\text{identified as faulty by quick test})}$

$= \dfrac{0.92 \times 0.06}{0.92 \times 0.06 + 0.08 \times 0.95}$

$= 0.42073$

$= 0.421$ (3 s.f.)

</td></tr>
<tr><td>(vii)</td><td>

$\mathrm{P}(\text{correct identification}) = 0.08 \times 0.95 + 0.92 \times 0.94$

$= 0.9408$

</td></tr>
<tr><td>(viii)</td><td>Although from part (vii), the quick test is 94.08% accurate, from part (vi), out of the number of lights identified as faulty, 42.1% of them are actually not faulty but are discarded and that results in a waste of resources. Hence, the quick test is not worthwhile.</td></tr>
</table>

10	**Suggested Solutions**

(i)

Let X be the mass of a randomly chosen metal sphere (in grams).

$X \sim \mathrm{N}(20, 0.5^2)$

$\mathrm{P}(X > 20.2) = 0.34458 = 0.345 \quad (3\text{ s.f.})$

(ii)

Let Y be the mass of a randomly chosen **coated metal sphere** (in grams).

Let $Y = 1.1X$

Due to the coating, the mass of the sphere increased by 10%, therefore, $Y = \frac{110}{100}X = 1.1X$

$\mathrm{E}(Y) = 1.1 \times 20 = 22$

$\mathrm{Var}(Y) = 1.1^2 \times 0.5^2 = 0.3025$

$Y \sim \mathrm{N}(22, 0.3025)$

$\mathrm{P}(21.5 < Y < 22.45) = 0.61172 = 0.612$

(iii)

Let W be the mass of a randomly chosen metal bar (in grams).
Let μ and σ be the population mean and population standard deviation of the masses of metal bars.

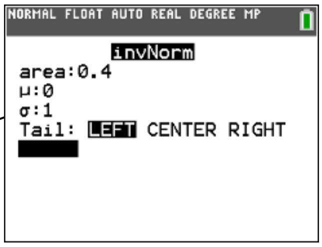
$W \sim \mathrm{N}\left(\mu, \sigma^2\right)$

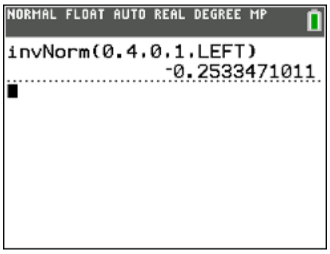
To solve for unknown μ and σ^2, we need to apply to concept of **standardisation**.

$\mathrm{P}(W > 12.2) = 0.6$

$\mathrm{P}\left(Z < \frac{12.2 - \mu}{\sigma}\right) = 0.4$

$\frac{12.2 - \mu}{\sigma} = -0.25335$



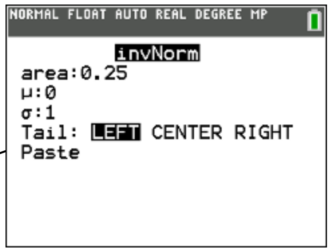


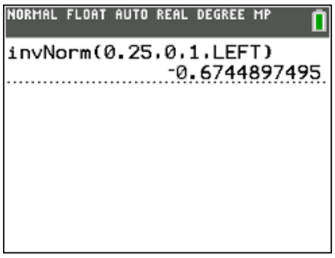
$\mu - 0.25335\sigma = 12.2 \;\ldots\ldots\; (1)$

$\mathrm{P}(W < 12) = 0.25$

$\mathrm{P}\left(Z < \frac{12 - \mu}{\sigma}\right) = 0.25$

$\frac{12 - \mu}{\sigma} = -0.67449$



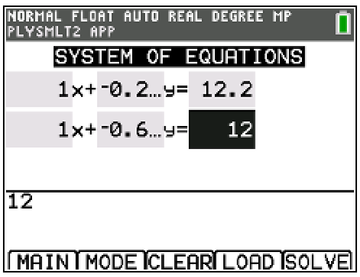


$\mu - 0.67449\sigma = 12 \;\ldots\ldots\; (2)$

Solving (1) & (2)

$\mu = 12.320 = 12.3 \;(3\text{s.f.})$

$\sigma = 0.47490 = 0.475 \;(3\text{s.f.})$



(iv) Let $S = \underbrace{Y_1 + Y_2}_{} + W$

mass of a randomly chosen metal bar

Note: total mass of **2 randomly chosen coated sphere** is $Y_1 + Y_2$

$\mathrm{E}(S) = 2\mathrm{E}(Y) + \mathrm{E}(W) = 44 + 12.320 = 56.32$

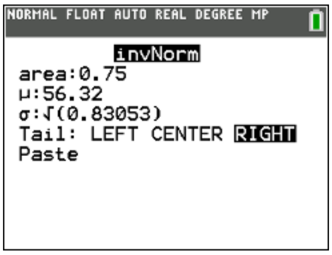
$\mathrm{Var}(S) = 2\mathrm{Var}(Y) + \mathrm{Var}(W) = 2 \times 0.3025 + 0.47490^2 = 0.83053$

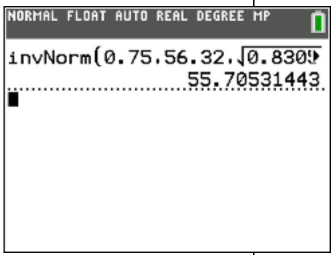
$S \sim \mathrm{N}(56.32, 0.83053)$

$\mathrm{P}(S > k) = 0.75$

"right" area

Using GC, $k = 55.705 = 55.7$ g (3s.f.)



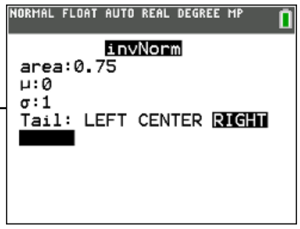


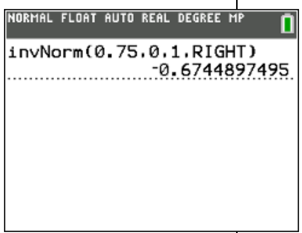
Learning point:
To find unknown constant k given the probability of 0.75, we can use the concept of taking "**inverse normal**"

Alternative method: Use standardisation

$S \sim \mathrm{N}(56.32, 0.83053)$

$\mathrm{P}\left(Z > \dfrac{k - 56.32}{\sqrt{0.83053}}\right) = 0.75$





$\dfrac{k - 56.32}{\sqrt{0.83053}} = -0.6744897495$

$k = 55.7053$

$k = 55.7$ (3 s.f)